

Assessment Methodology

SAKSHAM Baseline Assessment · Angul District, Odisha
Grade 5 & Grade 8 · Full census · July 2025

This note explains how the SAKSHAM assessment was designed, how it was run, and how a student's answers become the score out of 10 shown on each school report card. It is meant to be read alongside those report cards.

1. Purpose of the assessment

The SAKSHAM baseline was a diagnostic assessment. Its job was to show which foundational skills students had already secured and where they still needed help — not to rank or judge individual children.

- It was run as a census: every enrolled Grade 5 and Grade 8 student in the district's schools was meant to sit it, rather than a sample of them.
- The questions were pegged at foundational, earlier-grade levels (around the Grade 1 level) and mapped to the State's defined learning outcomes. A low score therefore points to specific building-block gaps that teaching can target.
- The same design was used across the whole district, so schools, clusters and blocks can be compared on a like-for-like basis.

2. What was assessed

- Grade 5 — four subjects: Odia, English, Mathematics, and Environmental Studies (EVS).
- Grade 8 — five subjects: Odia, English, Mathematics, Science, and Social Science.
- Format: paper-based multiple-choice questions with a single correct answer, recorded on OMR sheets for machine reading.
- Length: 15 questions per subject in Grade 5; 20 questions per subject in Grade 8.
- Time: about 30 minutes per subject in Grade 5, and 45 minutes in Grade 8.
- Short teacher, school and pupil questionnaires were collected alongside the tests to give context to the results.

The assessment ran over two days:

Day	Grade 5	Grade 8
Day 1	Odia, EVS	Odia, English, Science
Day 2	English, Maths	Maths, Social Science

3. How the assessment was conducted

- Students were assessed inside their own schools.
- External field investigators observed the testing to keep it fair and consistent from one school to the next.

- A buffer day let students who were absent sit the test later, so genuine absentees were not simply scored zero.
- For each subject, about twice as many questions were written as were finally needed. This let the questions be piloted and the weakest ones dropped before the real test, improving quality.

4. How a score is calculated

This is the core of the methodology: how a set of right and wrong answers becomes the score out of 10 on a report card. It happens in four steps.

4.1 Subject score (a percentage)

For each subject, a student's score is simply the share of questions they answered correctly:

$$\text{Subject score} = (\text{correct answers} \div \text{total questions}) \times 100$$

For example, a Grade 8 student who gets 14 of 20 Science questions right scores 70% in Science. A Grade 5 student who gets 10 of 15 Odia questions right scores about 67%.

4.2 Overall score (a percentage)

A student's overall score is their result across all subjects taken together. Because every subject has the same number of questions, this is the same as averaging the subject percentages.

School, cluster, block and district figures are then built up from these student scores — see Section 6.

4.3 From a percentage to a score out of 10

To make scores easy to read at a glance, the percentage is placed on a 0-to-10 scale using one simple rule:

$$\text{Score out of 10} = \text{percentage} \div 10, \text{ rounded to the nearest whole number}$$

So 62% becomes 6.2, which is shown as 6 out of 10; 78% becomes 8 out of 10. The report card shows this whole-number score, and the subject star ratings use the same idea. The exact percentage is kept in the downloadable data files, so no detail is lost.

4.4 Performance bands

Every score also falls into one of four colour-coded bands, decided by the percentage:

Band	Percentage	Out of 10
 Excelling	76 – 100%	8 – 10
 Developing	51 – 75%	6 – 7
 Needs support	26 – 50%	3 – 5
 Critical	0 – 25%	0 – 2

Bands give a quick sense of where a school stands without reading the exact number. The colour never carries meaning on its own — it always appears next to the band name and the score.

5. Learning-outcome (LO) mastery

Behind the subject scores sits a stricter, more detailed measure. Every question is linked to a specific learning outcome — a single skill or concept a student is expected to have.

- A student is counted as having mastered a learning outcome only if they answered every question under it correctly. Getting most of them right is not enough.
- For example, if a learning outcome has three questions and a student gets two right and one wrong, they have not mastered it.
- Because the bar is all-or-nothing, mastery percentages always look lower than subject scores. That is expected by design — it is a deliberately demanding test of whether a skill is truly secure.
- At school level, LO mastery is the share of students who mastered each outcome. Cluster and block figures are the average of the school-level figures.

6. How results roll up

Results are reported at five levels, each built from the one below it:

Student › School › Cluster › Block › District

- A school's score is the average of its students' scores.
- A cluster's or block's score is the average of the school scores within it — and here every school counts equally, whatever its size. A school with 4 students carries the same weight as one with 80.
- This is deliberate: it stops a few large schools from dominating the district picture. It does mean block and cluster averages describe schools, not individual students.

7. Data quality and responsible use

- Each answer is scored 1 for correct and 0 for wrong or blank.
- Where a school sat only one of the two days, the missing subjects are marked “not available” rather than counted as zero, and are left out of those averages. These are not errors.
- Schools with very low participation on a day (fewer than five students) are flagged so their figures are read with care.
- The assessment was built to give school-level insight, not to rank or judge individual students. Student-level records are kept anonymised and are meant to guide support — never punishment.

8. In short

A school shown as 6 out of 10 means its students, on average, answered a little over 60% of the questions correctly — which places it in the Developing band. The subject scores and learning-outcome details then show where, exactly, to focus effort.

